



Qualcomm Linux AI/ML Guide

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1 AI/ML documentation

Deploy pre-trained, deep learning models on Qualcomm hardware accelerators, such as Neural Processing Unit (NPU), Graphic Processing Unit (GPU), and Central Processing Unit (CPU).

Qualcomm Linux offers software development kits (SDKs), APIs, sample applications, development tools, and third-party framework support to ease AI application development.

2 AI software overview and architecture

Qualcomm® Linux AI software allows you to optimally deploy pre-trained, deep learning models on Qualcomm hardware accelerators, such as neural processing unit (NPU), graphic processing unit (GPU), and central processing unit (CPU). Qualcomm Linux contains software development kits (SDKs), APIs, sample applications, development tools, and third-party frameworks support, such as GStreamer and TFLite, for application development.

Multimedia AI applications use the Qualcomm IM SDK GStreamer Plugin APIs to take advantage of Qualcomm AI Runtime SDK APIs that provide lower-level hardware acceleration for AI use cases, as shown in the following figure:

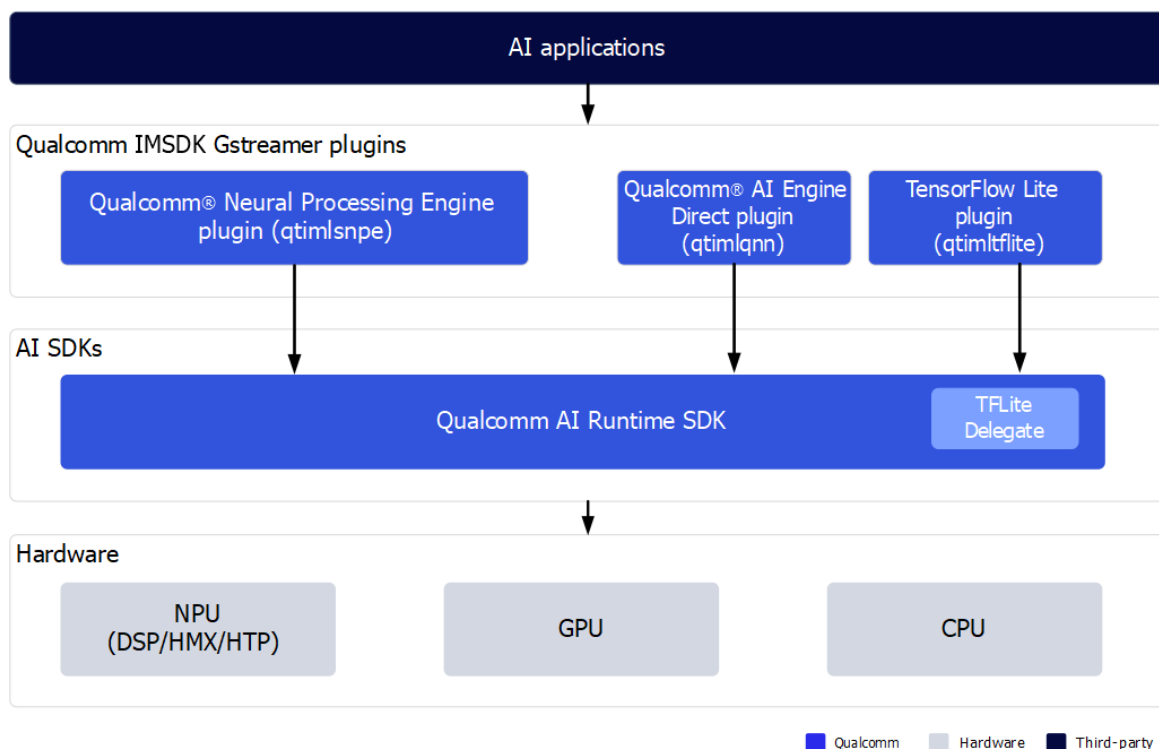


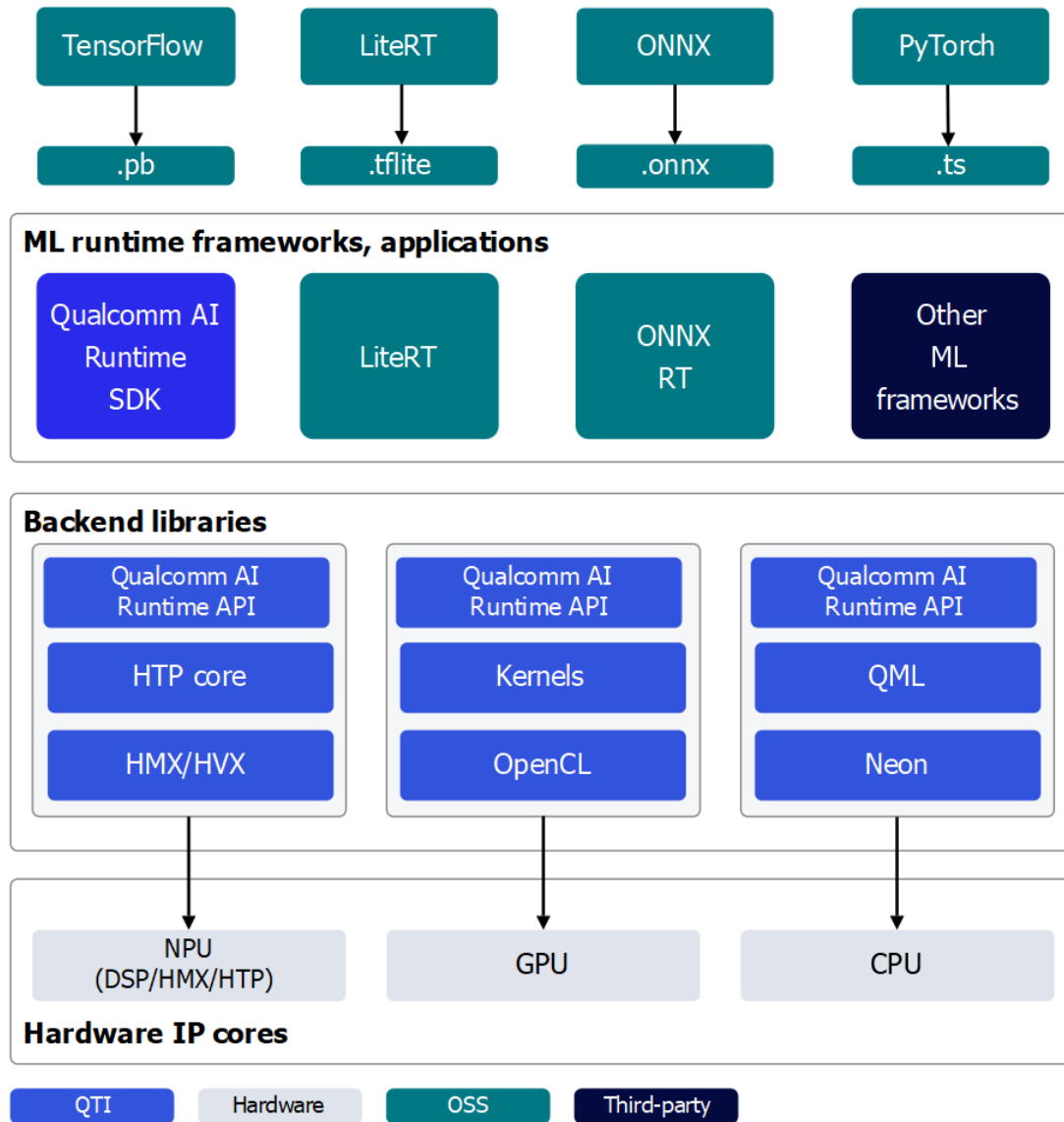
Figure1 Top-level AI hardware and software blocks of the Qualcomm Linux AI stack

The key components of the Qualcomm Linux AI software are as follows:

- **AI applications:** Sample applications based on Gstreamer that you can [customize](#).
- **GStreamer plugins:** Qualcomm Linux software offers GStreamer-based, machine learning plugins to accelerate AI inference using tools such as, LiteRT and Qualcomm AI Runtime SDK, along with GStreamer plugins for pre- and postprocessing.
- **Qualcomm AI Runtime SDK (QAI RT):** QAI RT provides tools and libraries to optimally accelerate AI models on multiple hardware accelerators.
- Qualcomm System-on-Chip (SoC) offer three hardware cores for AI loads.
 - **Neural Processing Unit (NPU):** The NPU, also referred to as Qualcomm® Hexagon™ Tensor Processor (HTP) or DSP/HMX, is suitable for running AI workloads with low-power and high-performance. To get optimized performance, quantize pre-trained models to one of the supported precisions.
 - **Graphics Processing Unit (GPU):** Qualcomm® Adreno™ GPU is suitable for running AI workloads with medium-power and medium-performance. AI workloads are accelerated with OpenCL kernels. You can also use the GPU to accelerate model pre- and postprocessing.
 - **Central Processing Unit (CPU):** Use AI inferencing on the CPU to benchmark model accuracy or performance against other hardware accelerators. You can also use the CPU to run model pre- and postprocessing.

2.1 Architecture

The following image shows available AI runtimes, such as ONNX runtime and LiteRT, the available hardware accelerators, such as the NPU, GPU, and CPU, and the backend libraries for each accelerator.



AI hardware accelerators

You can accelerate AI workflows using the following hardware cores:

- **Qualcomm® Hexagon™ Tensor Processor (HTP):** The HTP, also known as NPU, DSP, or HMX, is suitable to run AI workloads with low-power and high-performance. To get optimized performance, quantize pre-trained models to one of the supported precisions.
- **Qualcomm® Adreno™ GPU:** Suitable to run AI workloads with medium-power and medium-performance. AI workloads are accelerated with OpenCL kernels. You can also use the GPU to accelerate model pre- and postprocessing.
- **Qualcomm® Kryo™ CPU:** Use AI inferencing on the CPU to benchmark model accuracy and performance against other hardware accelerators. You can also use the CPU to run model pre- and postprocessing.

AI software

Choose from the following software to deploy AI workloads. You must convert pre-trained models into an executable format with the SDK before running them, except for TFLite models. Pre-trained models (with the exception of LiteRT models) need to be converted to an executable format with the selected AI Stack SDK before running them. The TFLite Delegate allows you to directly run LiteRT models.

- **[Qualcomm AI Runtime SDK \(QAIRT\)](#):** QAIRT SDK is a unified software package that integrates Qualcomm AI tools, including AI Engine Direct and Neural Processing SDK. It enables developers to port and deploy AI models on Qualcomm hardware accelerators (CPU, GPU, NPU) and supports popular frameworks such as TensorFlow, PyTorch, LiteRT, and ONNX for efficient model execution across Qualcomm platforms.
- **[AI Model Efficiency Toolkit \(AIMET\)](#):** Open-source library to optimize (compressing and quantizing) trained neural network models. This is a complex SDK designed to generate optimized quantized models.

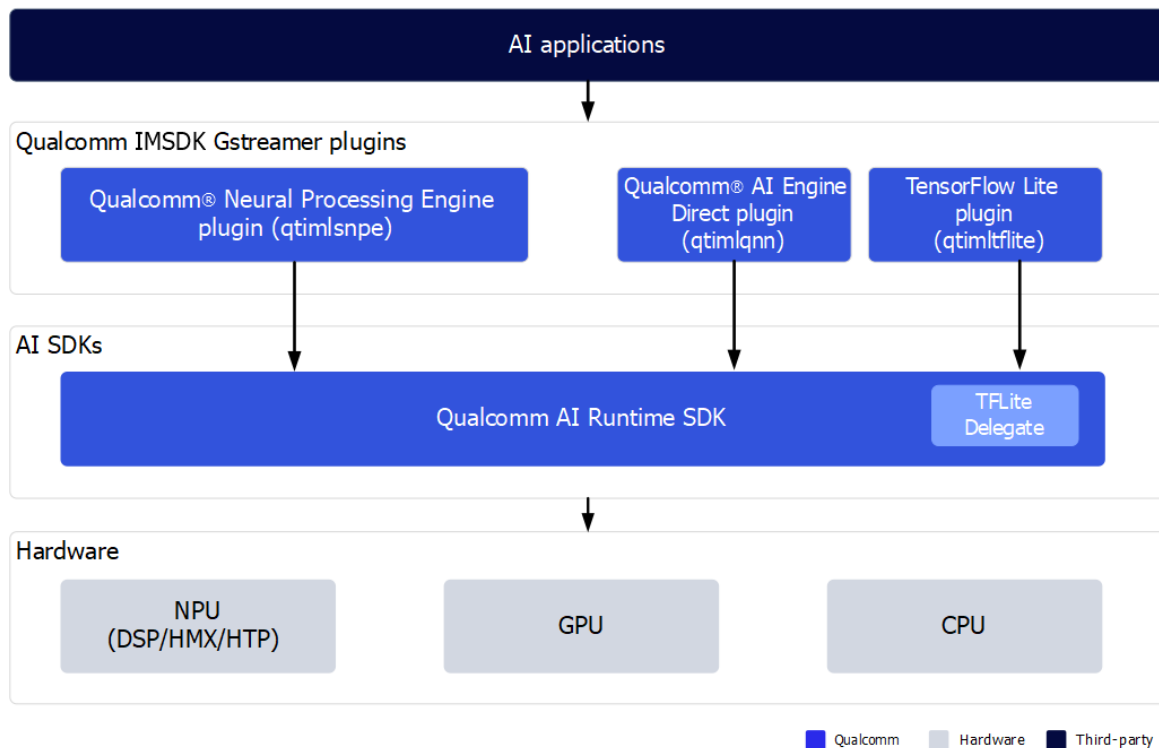
2.2 Next steps

- [AI Hub](#)
- [Qualcomm AI Model Efficiency Toolkit](#)
- [Qualcomm AI Runtime SDK](#)
- [Qualcomm Intelligent Multimedia SDK](#)

3 GStreamer ML plugin and AI SDK APIs

Qualcomm Linux provides [Qualcomm® Intelligent Multimedia SDK \(IM SDK\) GStreamer plugin APIs](#) to interface with Qualcomm AI Runtime SDK (QAIRT) APIs, to optimally run deep learning models on hardware modules such as neural processing unit (NPU), graphics processing unit (GPU), and central processing unit (CPU).

The following software block diagram shows how AI applications can call Qualcomm IM SDK GStreamer Plugin APIs (`qtimlsnpe`, `qtimlqnn`, `qtimltflite`) to interface with and access QAIRT APIs to use a specific hardware accelerator.



3.1 GStreamer Plugin APIs

Three GStreamer ML plugin APIs support the QAI RT SDK and TFLite, which provides flexibility for you to choose the right combination for your needs.

Qualcomm IM SDK plugin for Qualcomm Neural Processing Engine (qtimlsnpe)	Uses Qualcomm AI Runtime APIs to load and execute models. Choose this plugin for quick prototyping and high-level API support.
Qualcomm IM SDK plugin for Qualcomm AI Engine Direct plugin (qtimlqnn)	Uses Qualcomm AI Runtime APIs, which provide low-level, unified API and improved performance to optimize and execute network models on the desired hardware accelerator. Choose this plugin for advanced graph execution options and optimizations.
Qualcomm IM SDK plugin for LiteRT (qtimlflite)	Accelerates LiteRT models directly using Qualcomm AI Runtime APIs to load and execute models. Choose this plugin to directly run LiteRT models, without conversion.

3.2 Qualcomm AI Runtime SDK APIs

Use the [QAI RT](#) APIs to develop AI applications. The C, C++, and Java APIs support heterogeneous computing, system-level configurations, and direct AI workloads to all accelerator cores. QAI RT provides low-level, highly customizable APIs to target a specific accelerator core or delegate workloads from popular runtimes including the LiteRT and ONNX runtimes.

4 Hardware-accelerated sample applications for AI

Qualcomm Linux provides sample AI/ML applications that show the AI capabilities of the Qualcomm Linux platform. The following image shows the workflow to run these AI sample apps, from booting the device, downloading the models, transferring the models to the device to running the sample apps on the target device.

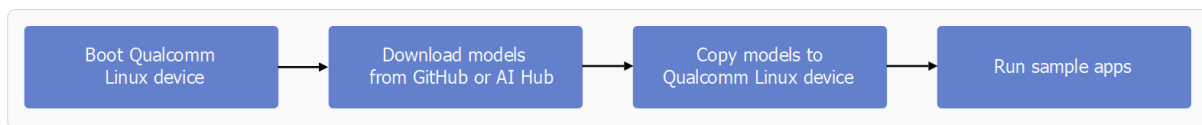


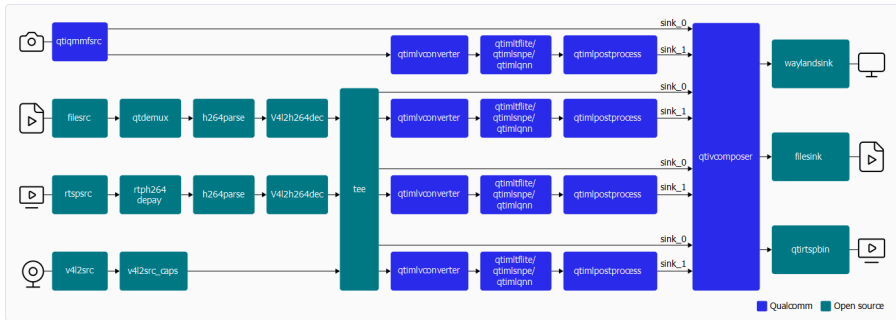
Figure1 Workflow for using the sample applications

The following AI sample applications are part of Qualcomm IM SDK. For the complete list of supported sample applications, see [Sample applications](#) in the Qualcomm IM SDK documentation.

4.1 AI processing for object detection

The [gst-ai-object-detection](#) application allows you to detect objects within images and videos. The use cases show the execution of [YOLOv5](#), [YOLOv8](#), and [YOLOX](#) using the Qualcomm Neural Processing SDK runtime, YOLOv8 using Qualcomm AI Engine direct, and YOLOv5 and YOLOv8 using LiteRT.

The following figure shows the pipeline, which receives the input from a live camera feed, file, or an RTSP stream, preprocesses it, runs inferences on AI hardware, and displays the results on the screen. For information about the plugins used in the pipeline flow, see [object detection pipeline flow](#).



4.2 AI processing for multistream-inference

The [gst-ai-multistream-inference](#) application shows AI inference (object detection and classification) on up to 32 input streams from camera, file, or RTSP stream.

The following figure shows the pipeline, which receives several input streams, preprocesses them, runs AI inferences, combines the streams, and merges them all into a single video output.

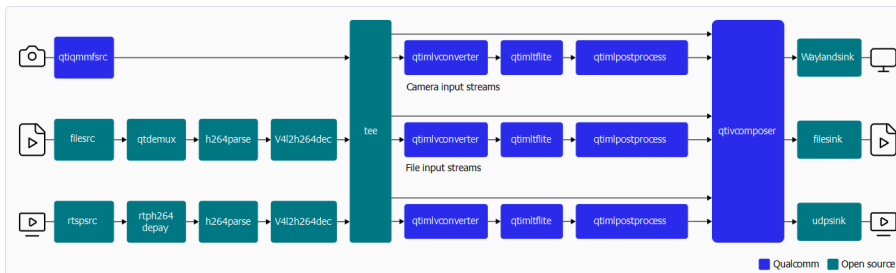
The maximum number of input streams supported on each SoC are follows:

- QCS6490: 8
- Qualcomm Dragonwing IQ-8275: 16
- Qualcomm Dragonwing IQ-9075: 32

Note: IQ-8275 is not supported on Ubuntu Server.

The output is displayed on an HDMI display, saved as an H.264 encoded MP4 file, or converted to an RTSP stream.

For information about the plugins used in this pipeline, see [multistream-inference pipeline flow](#).



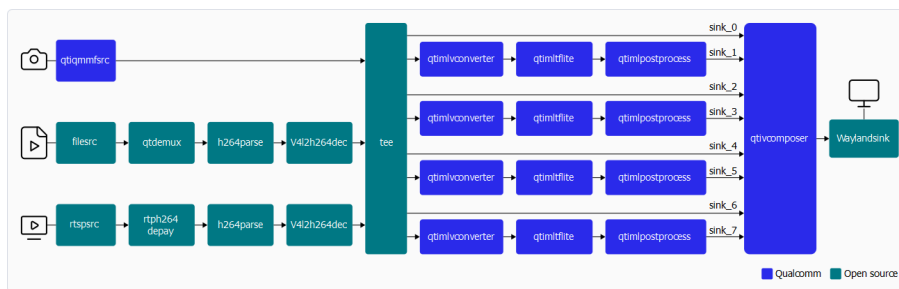
4.3 Parallel AI inference

The [gst-ai-parallel-inference](#) application allows you to perform object detection, object classification, pose detection, and image segmentation on an input stream from different sources, such as camera, file, or an RTSP network. The use cases implement the LiteRT models for object detection, image segmentation, classification, and pose detection.

Note: This application isn't supported on IQ-8275.

The following figure shows the pipeline, which receives input streams from a camera, file, or an RTSP stream, performs the parallel inferencing for the four use cases, and displays the results side by side on the screen.

For information about the plugins used in the pipeline flow, see [parallel inference pipeline flow](#).

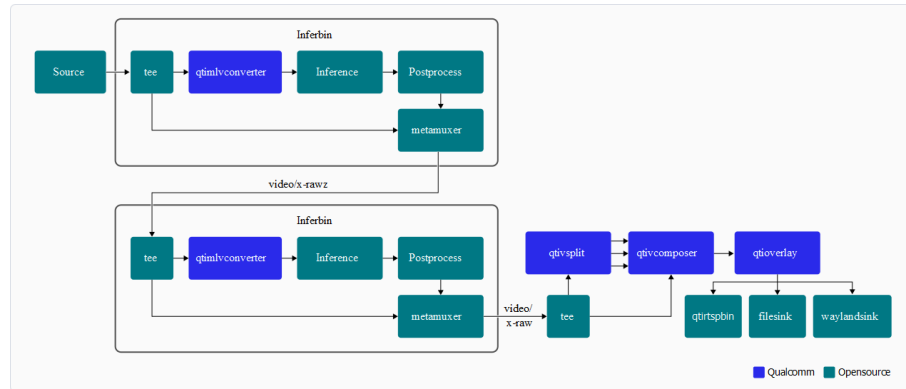


4.4 Daisychain of object detection and pose estimation

The [gst-ai-daisychain-detection-pose](#) application allows you to perform cascaded object detection and pose detection with a camera, file source, or an RTSP stream. The use cases involve detecting objects and estimating the body poses of the subject in an image or a video.

The following figure show the application workflow, which receives the source, postprocesses it, runs inferences on AI hardware, and displays the results on the screen.

For information about the plugins used in the pipeline flow, see [daisychain detection and pose pipeline flow](#).



5 Develop an AI application

Developing an AI application for Qualcomm Linux involves compiling and optimizing models and AI frameworks to run efficiently on Qualcomm hardware.

This process involves the following steps:

1. Selecting an AI model and framework.
2. Compiling and optimizing the model to run on Qualcomm hardware.
3. Building an application that integrates the optimized model.

For more information about compiling AI models and developing AI/ML reference applications, see the [AI/ML Developer Workflow](#).

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